

1. Executive Summary

Project Name: OurClass: Context-Aware Smart Classroom & Kiosk Engine

Target Audience: Educational Institutions, Faculty, and Students

Primary Objective: Transform student smartphones during college hours into distraction-free Digital Study Terminals using AI and Android Kiosk Architecture.

2. Problem Statement

- **Smartphone Distractions:** During lectures, students often engage in social media, gaming, and non-academic content on their devices.
- **Ineffectiveness of Total Bans:** Complete phone bans are impractical and eliminate the educational potential of modern mobile devices.
- **Note-Taking Challenges:** Multi-tasking between listening to lectures and taking detailed notes leads to reduced comprehension.

3. Proposed Solution & Key Modules

A. Mobile Kiosk & Dynamic Locking System

- **Classroom Geofencing / Beacon Trigger:** Automatically activates secure mode when connected to campus Wi-Fi or entering designated classroom boundaries.
- **App Blocker (Lock Task Mode):** Blocks social media, messaging, games, and unauthorized incoming/outgoing calls during active class duration.
- **Automatic Release:** Restores full smartphone capabilities instantly upon session completion or faculty authorization.

B. Faculty Control Panel (Admin / Host)

- **Host Privileges:** Only registered faculty members can initiate or end classroom sessions.
- **Subject Mapping:** Automatically displays relevant study materials, notes, and interactive quizzes based on the current subject (e.g., Commerce, Computer Science).

C. Integrated Microsoft AI Stack

- **Real-time Speech-to-Notes (Azure Speech Services):** Transcribes live lecture audio into structured digital notes on student screens in real-time.
- **Guarded Contextual Q&A (Azure OpenAI):** Answers subject-specific doubts while enforcing strict guardrails against off-topic queries.

4. System Architecture & Tech Stack

Component	Technology Stack	Functional Description
Mobile Application	Flutter / React Native	Cross-platform app supporting dual roles (Faculty Host & Student Member).
Kiosk / Locking Engine	Android Native Kiosk API (<code>`startLockTask()`</code>)	Restricts system access and hides non-academic app notifications.
Cloud & Backend	Node.js / Python FastAPI	Handles session state, user authentication, and API routing.
AI Engine	Microsoft Azure OpenAI & Speech Services	Powers automated transcription, note generation, and guarded Q&A.

5. Key Value Proposition & Impact

- **Enhanced Academic Focus:** Ensures a distraction-free learning environment without removing modern technology.
- **AI-Assisted Learning:** Real-time transcription and subject-bound AI assistant boost student comprehension.
- **Institutional Control:** Empowers faculty and institution admins with effective classroom management tools.